

- 2 (a) The hormone glucagon is an example of a cell signalling molecule. Table 2.1 lists the main events that occur when the blood glucose concentration decreases below the set point.

The events are **not** listed in the correct order.

Table 2.1

event	description of event
A	adenylyl cyclase enzyme is activated
B	cyclic AMP activates an enzyme cascade
C	glycogen stored in liver cells is broken down to glucose
D	blood glucose concentration increases
E	glucagon is secreted by α cells in the pancreas
F	conformational change to glucagon receptor causes G-protein activation
G	active adenylyl cyclase acts on ATP to produce second messenger
H	glucagon signal is amplified
I	glucose diffuses out of liver cells through GLUT transporter proteins
J	glucagon binds to receptors in the cell surface membranes of liver cells
K	cyclic AMP is formed

Complete Table 2.2 to show the correct order in which these events occur.

Three of the events have already been placed in their correct order.

Table 2.2

correct order	letter of event
1	E
2	
3	
4	
5	
6	K
7	
8	
9	
10	
11	D

(b) An investigation was carried out to measure the rate at which glucose is provided for respiration from three different sources of glucose:

- a meal
- glycogenolysis – the breakdown of glycogen
- gluconeogenesis – production of glucose from non-carbohydrate molecules.

After a person ate a meal, the rates at which glucose was provided for respiration from the three different sources were measured at regular intervals over a 24-hour period. During this period, no food was eaten.

Fig. 2.1 shows the results of this investigation.

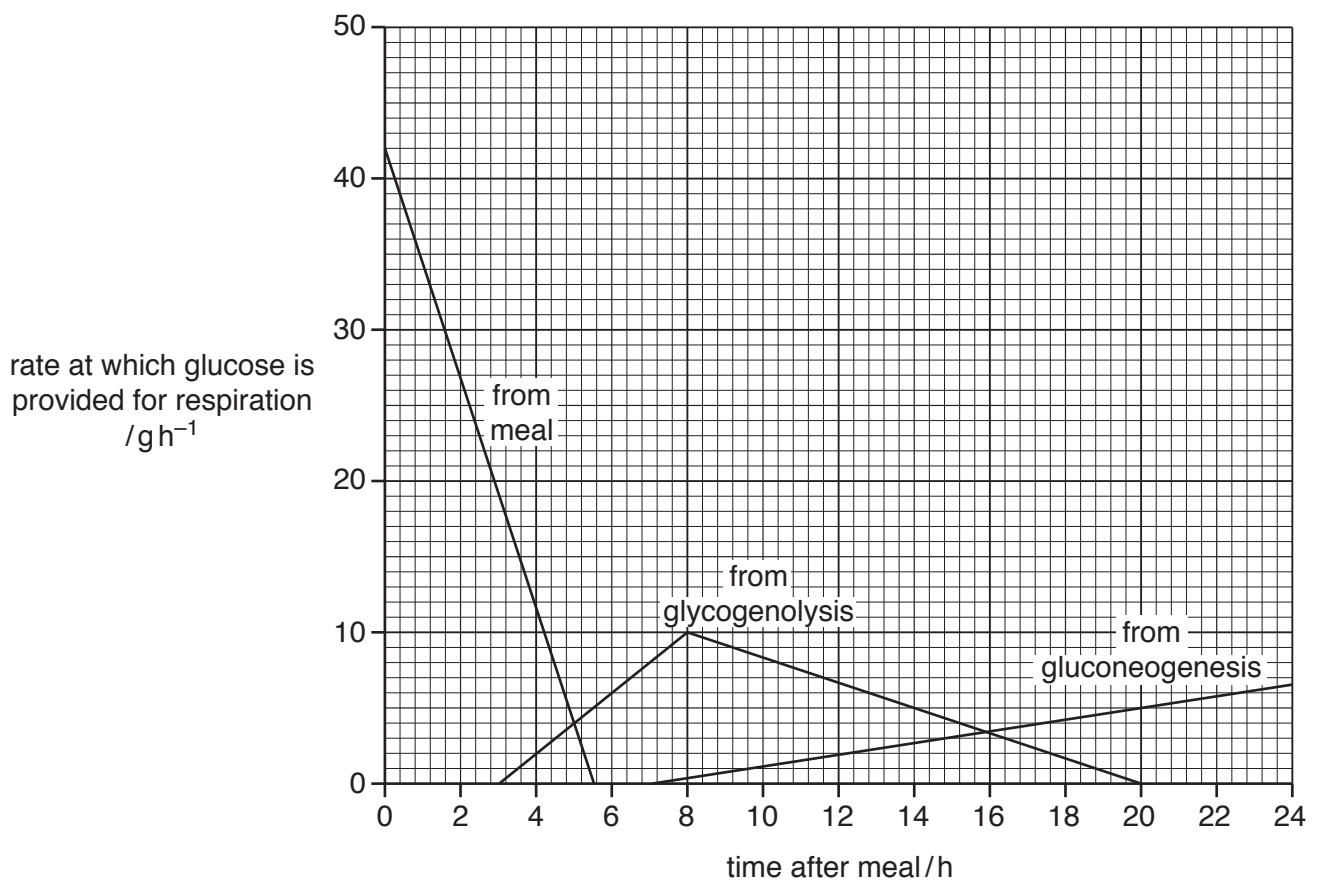


Fig. 2.1

- (i) State the time after the meal when the rate at which glucose was provided from the meal for respiration was the same as the rate at which glucose was provided from glycogenolysis for respiration.

..... [1]

- (ii) State the first time after the meal when **all** of the glucose for respiration was provided by gluconeogenesis.

..... [1]

- (iii) Name the homeostatic mechanism by which blood glucose concentration is maintained at a set point.

..... [1]

- (iv) In humans, carbohydrates such as glucose are not the only respiratory substrates.

Name **two** non-carbohydrate respiratory substrates in humans.

.....

..... [2]

[Total: 9]